Chapter 1

A Runaway Reef

THE YEAR 1866 was marked by a bizarre development, an unexplained and downright inexplicable phenomenon that surely no one has forgotten. Without getting into those rumors that upset civilians in the seaports and deranged the public mind even far inland, it must be said that professional seamen were especially alarmed. Traders, shipowners, captains of vessels, skippers, and master mariners from Europe and America, naval officers from every country, and at their heels the various national governments on these two continents, were all extremely disturbed by the business.

In essence, over a period of time several ships had encountered “an enormous thing” at sea, a long spindle-shaped object, sometimes giving off a phosphorescent glow, infinitely bigger and faster than any whale.

The relevant data on this apparition, as recorded in various logbooks, agreed pretty closely as to the structure of the object or creature in question, its unprecedented speed of movement, its startling locomotive power, and the unique vitality with which it seemed to be gifted. If it was a cetacean, it exceeded in bulk any whale previously classified by science. No naturalist, neither Cuvier nor Lacépède, neither Professor Dumeril nor Professor de Quatrefages, would have accepted the existence of such a monster sight unseen—specifically, unseen by their own scientific eyes.

Striking an average of observations taken at different times—rejecting those timid estimates that gave the object a length of 200 feet, and ignoring those exaggerated views that saw it as a mile wide and three long—you could

still assert that this phenomenal creature greatly exceeded the dimensions of anything then known to ichthyologists, if it existed at all.

Now then, it did exist, this was an undeniable fact; and since the human mind dotes on objects of wonder, you can understand the worldwide excitement caused by this unearthly apparition. As for relegating it to the realm of fiction, that charge had to be dropped.

In essence, on July 20, 1866, the steamer Governor Higginson, from the Calcutta & Burnach Steam Navigation Co., encountered this moving mass five miles off the eastern shores of Australia.

Captain Baker at first thought he was in the presence of an unknown reef; he was even about to fix its exact position when two waterspouts shot out of this inexplicable object and sprang hissing into the air some 150 feet. So, unless this reef was subject to the intermittent eruptions of a geyser, the Governor Higginson had fair and honest dealings with some aquatic mammal, until then unknown, that could spurt from its blowholes waterspouts mixed with air and steam.

Similar events were likewise observed in Pacific seas, on July 23 of the same year, by the Christopher Columbus from the West India & Pacific Steam Navigation Co. Consequently, this extraordinary cetacean could transfer itself from one locality to another with startling swiftness, since within an interval of just three days, the Governor Higginson and the Christopher Columbus had observed it at two positions on the charts separated by a distance of more than 700 nautical leagues.

Fifteen days later and 2,000 leagues farther, the Helvetia from the Compagnie Nationale and the Shannon from the Royal Mail line, running on opposite tacks in that part of the Atlantic lying between the United States and Europe, respectively signaled each other that the monster had been sighted in latitude 42 degrees 15′ north and longitude 60 degrees 35′ west of the meridian of Greenwich. From their simultaneous observations, they were able to estimate the mammal’s minimum length at more than 350 English feet;* this was because both the Shannon and the Helvetia were of smaller dimensions,

although each measured 100 meters stem to stern. Now then, the biggest whales, those rorqual whales that frequent the waterways of the Aleutian Islands, have never exceeded a length of 56 meters—if they reach even that.

*Author's Note: About 106 meters. An English foot is only 30.4 centimeters.

One after another, reports arrived that would profoundly affect public opinion: new observations taken by the transatlantic liner Pereire, the Inman line's Etna running afoul of the monster, an official report drawn up by officers on the French frigate Normandy, dead-earnest reckonings obtained by the general staff of Commodore Fitz-James aboard the Lord Clyde. In lighthearted countries, people joked about this phenomenon, but such serious, practical countries as England, America, and Germany were deeply concerned.

In every big city the monster was the latest rage; they sang about it in the coffee houses, they ridiculed it in the newspapers, they dramatized it in the theaters. The tabloids found it a fine opportunity for hatching all sorts of hoaxes. In those newspapers short of copy, you saw the reappearance of every gigantic imaginary creature, from "Moby Dick," that dreadful white whale from the High Arctic regions, to the stupendous kraken whose tentacles could entwine a 500-ton craft and drag it into the ocean depths. They even reprinted reports from ancient times: the views of Aristotle and Pliny accepting the existence of such monsters, then the Norwegian stories of Bishop Pontoppidan, the narratives of Paul Egede, and finally the reports of Captain Harrington—whose good faith is above suspicion—in which he claims he saw, while aboard the Castilian in 1857, one of those enormous serpents that, until then, had frequented only the seas of France's old extremist newspaper, The Constitutionalist.

An interminable debate then broke out between believers and skeptics in the scholarly societies and scientific journals. The "monster question" inflamed all minds. During this memorable campaign, journalists making a profession of science battled with those making a profession of wit, spilling

waves of ink and some of them even two or three drops of blood, since they went from sea serpents to the most offensive personal remarks.

For six months the war seesawed. With inexhaustible zest, the popular press took potshots at feature articles from the Geographic Institute of Brazil, the Royal Academy of Science in Berlin, the British Association, the Smithsonian Institution in Washington, D.C., at discussions in The Indian Archipelago, in Cosmos published by Father Moigno, in Petermann's Mittheilungen,* and at scientific chronicles in the great French and foreign newspapers. When the monster's detractors cited a saying by the botanist Linnaeus that "nature doesn't make leaps," witty writers in the popular periodicals parodied it, maintaining in essence that "nature doesn't make lunatics," and ordering their contemporaries never to give the lie to nature by believing in krakens, sea serpents, "Moby Dicks," and other all-out efforts from drunken seamen. Finally, in a much-feared satirical journal, an article by its most popular columnist finished off the monster for good, spurning it in the style of Hippolytus repulsing the amorous advances of his stepmother Phaedra, and giving the creature its quietus amid a universal burst of laughter. Wit had defeated science.

*German: "Bulletin." Ed.

During the first months of the year 1867, the question seemed to be buried, and it didn't seem due for resurrection, when new facts were brought to the public's attention. But now it was no longer an issue of a scientific problem to be solved, but a quite real and serious danger to be avoided. The question took an entirely new turn. The monster again became an islet, rock, or reef, but a runaway reef, unfixed and elusive.

On March 5, 1867, the Moravian from the Montreal Ocean Co., lying during the night in latitude 27 degrees 30′ and longitude 72 degrees 15′, ran its starboard quarter afoul of a rock marked on no charts of these waterways. Under the combined efforts of wind and 400-horsepower steam, it was traveling at a speed of thirteen knots. Without the high quality of its hull, the

Moravian would surely have split open from this collision and gone down together with those 237 passengers it was bringing back from Canada.

This accident happened around five o'clock in the morning, just as day was beginning to break. The officers on watch rushed to the craft's stern. They examined the ocean with the most scrupulous care. They saw nothing except a strong eddy breaking three cable lengths out, as if those sheets of water had been violently churned. The site's exact bearings were taken, and the Moravian continued on course apparently undamaged. Had it run afoul of an underwater rock or the wreckage of some enormous derelict ship? They were unable to say. But when they examined its undersides in the service yard, they discovered that part of its keel had been smashed.

This occurrence, extremely serious in itself, might perhaps have been forgotten like so many others, if three weeks later it hadn't been reenacted under identical conditions. Only, thanks to the nationality of the ship victimized by this new ramming, and thanks to the reputation of the company to which this ship belonged, the event caused an immense uproar.

No one is unaware of the name of that famous English shipowner, Cunard. In 1840 this shrewd industrialist founded a postal service between Liverpool and Halifax, featuring three wooden ships with 400-horsepower paddle wheels and a burden of 1,162 metric tons. Eight years later, the company's assets were increased by four 650-horsepower ships at 1,820 metric tons, and in two more years, by two other vessels of still greater power and tonnage. In 1853 the Cunard Co., whose mail-carrying charter had just been renewed, successively added to its assets the Arabia, the Persia, the China, the Scotia, the Java, and the Russia, all ships of top speed and, after the Great Eastern, the biggest ever to plow the seas. So in 1867 this company owned twelve ships, eight with paddle wheels and four with propellers.

If I give these highly condensed details, it is so everyone can fully understand the importance of this maritime transportation company, known the world over for its shrewd management. No transoceanic navigational undertaking has been conducted with more ability, no business dealings have

been crowned with greater success. In twenty-six years Cunard ships have made 2,000 Atlantic crossings without so much as a voyage canceled, a delay recorded, a man, a craft, or even a letter lost. Accordingly, despite strong competition from France, passengers still choose the Cunard line in preference to all others, as can be seen in a recent survey of official documents. Given this, no one will be astonished at the uproar provoked by this accident involving one of its finest steamers.

On April 13, 1867, with a smooth sea and a moderate breeze, the Scotia lay in longitude 15 degrees 12′ and latitude 45 degrees 37′. It was traveling at a speed of 13.43 knots under the thrust of its 1,000-horsepower engines. Its paddle wheels were churning the sea with perfect steadiness. It was then drawing 6.7 meters of water and displacing 6,624 cubic meters.

At 4:17 in the afternoon, during a high tea for passengers gathered in the main lounge, a collision occurred, scarcely noticeable on the whole, affecting the Scotia's hull in that quarter a little astern of its port paddle wheel.

The Scotia hadn't run afoul of something, it had been fouled, and by a cutting or perforating instrument rather than a blunt one. This encounter seemed so minor that nobody on board would have been disturbed by it, had it not been for the shouts of crewmen in the hold, who climbed on deck yelling:

"We're sinking! We're sinking!"

At first the passengers were quite frightened, but Captain Anderson hastened to reassure them. In fact, there could be no immediate danger. Divided into seven compartments by watertight bulkheads, the Scotia could brave any leak with impunity.

Captain Anderson immediately made his way into the hold. He discovered that the fifth compartment had been invaded by the sea, and the speed of this invasion proved that the leak was considerable. Fortunately this compartment didn't contain the boilers, because their furnaces would have been abruptly extinguished.

Captain Anderson called an immediate halt, and one of his sailors dived down to assess the damage. Within moments they had located a hole two

meters in width on the steamer's underside. Such a leak could not be patched, and with its paddle wheels half swamped, the Scotia had no choice but to continue its voyage. By then it lay 300 miles from Cape Clear, and after three days of delay that filled Liverpool with acute anxiety, it entered the company docks.

The engineers then proceeded to inspect the Scotia, which had been put in dry dock. They couldn't believe their eyes. Two and a half meters below its waterline, there gaped a symmetrical gash in the shape of an isosceles triangle. This breach in the sheet iron was so perfectly formed, no punch could have done a cleaner job of it. Consequently, it must have been produced by a perforating tool of uncommon toughness—plus, after being launched with prodigious power and then piercing four centimeters of sheet iron, this tool had needed to withdraw itself by a backward motion truly inexplicable.

This was the last straw, and it resulted in arousing public passions all over again. Indeed, from this moment on, any maritime casualty without an established cause was charged to the monster's account. This outrageous animal had to shoulder responsibility for all derelict vessels, whose numbers are unfortunately considerable, since out of those 3,000 ships whose losses are recorded annually at the marine insurance bureau, the figure for steam or sailing ships supposedly lost with all hands, in the absence of any news, amounts to at least 200!

Now then, justly or unjustly, it was the "monster" who stood accused of their disappearance; and since, thanks to it, travel between the various continents had become more and more dangerous, the public spoke up and demanded straight out that, at all cost, the seas be purged of this fearsome cetacean.

Chapter 2

The Pros and Cons

DURING THE PERIOD in which these developments were occurring, I had returned from a scientific undertaking organized to explore the Nebraska badlands in the United States. In my capacity as Assistant Professor at the Paris Museum of Natural History, I had been attached to this expedition by the French government. After spending six months in Nebraska, I arrived in New York laden with valuable collections near the end of March. My departure for France was set for early May. In the meantime, then, I was busy classifying my mineralogical, botanical, and zoological treasures when that incident took place with the Scotia.

I was perfectly abreast of this question, which was the big news of the day, and how could I not have been? I had read and reread every American and European newspaper without being any farther along. This mystery puzzled me. Finding it impossible to form any views, I drifted from one extreme to the other. Something was out there, that much was certain, and any doubting Thomas was invited to place his finger on the Scotia's wound.

When I arrived in New York, the question was at the boiling point. The hypothesis of a drifting islet or an elusive reef, put forward by people not quite in their right minds, was completely eliminated. And indeed, unless this reef had an engine in its belly, how could it move about with such prodigious speed?

Also discredited was the idea of a floating hull or some other enormous wreckage, and again because of this speed of movement.

So only two possible solutions to the question were left, creating two very distinct groups of supporters: on one side, those favoring a monster of colossal strength; on the other, those favoring an "underwater boat" of tremendous motor power.

Now then, although the latter hypothesis was completely admissible, it couldn't stand up to inquiries conducted in both the New World and the Old.

That a private individual had such a mechanism at his disposal was less than probable. Where and when had he built it, and how could he have built it in secret?

Only some government could own such an engine of destruction, and in these disaster-filled times, when men tax their ingenuity to build increasingly powerful aggressive weapons, it was possible that, unknown to the rest of the world, some nation could have been testing such a fearsome machine. The Chassepot rifle led to the torpedo, and the torpedo has led to this underwater battering ram, which in turn will lead to the world putting its foot down. At least I hope it will.

But this hypothesis of a war machine collapsed in the face of formal denials from the various governments. Since the public interest was at stake and transoceanic travel was suffering, the sincerity of these governments could not be doubted. Besides, how could the assembly of this underwater boat have escaped public notice? Keeping a secret under such circumstances would be difficult enough for an individual, and certainly impossible for a nation whose every move is under constant surveillance by rival powers.

So, after inquiries conducted in England, France, Russia, Prussia, Spain, Italy, America, and even Turkey, the hypothesis of an underwater Monitor was ultimately rejected.

And so the monster surfaced again, despite the endless witticisms heaped on it by the popular press, and the human imagination soon got caught up in the most ridiculous ichthyological fantasies.

After I arrived in New York, several people did me the honor of consulting me on the phenomenon in question. In France I had published a two-volume work, in quarto, entitled The Mysteries of the Great Ocean Depths. Well received in scholarly circles, this book had established me as a specialist in this pretty obscure field of natural history. My views were in demand. As long as I could deny the reality of the business, I confined myself to a flat “no comment.” But soon, pinned to the wall, I had to explain myself straight out. And in this vein, “the honorable Pierre Aronnax, Professor at the Paris

Museum," was summoned by The New York Herald to formulate his views no matter what.

I complied. Since I could no longer hold my tongue, I let it wag. I discussed the question in its every aspect, both political and scientific, and this is an excerpt from the well-padded article I published in the issue of April 30.

"Therefore," I wrote, "after examining these different hypotheses one by one, we are forced, every other supposition having been refuted, to accept the existence of an extremely powerful marine animal.

"The deepest parts of the ocean are totally unknown to us. No soundings have been able to reach them. What goes on in those distant depths? What creatures inhabit, or could inhabit, those regions twelve or fifteen miles beneath the surface of the water? What is the constitution of these animals? It's almost beyond conjecture.

"However, the solution to this problem submitted to me can take the form of a choice between two alternatives.

"Either we know every variety of creature populating our planet, or we do not.

"If we do not know every one of them, if nature still keeps ichthyological secrets from us, nothing is more admissible than to accept the existence of fish or cetaceans of new species or even new genera, animals with a basically 'cast-iron' constitution that inhabit strata beyond the reach of our soundings, and which some development or other, an urge or a whim if you prefer, can bring to the upper level of the ocean for long intervals.

"If, on the other hand, we do know every living species, we must look for the animal in question among those marine creatures already cataloged, and in this event I would be inclined to accept the existence of a giant narwhale.

"The common narwhale, or sea unicorn, often reaches a length of sixty feet. Increase its dimensions fivefold or even tenfold, then give this cetacean a strength in proportion to its size while enlarging its offensive weapons, and you have the animal we're looking for. It would have the proportions

determined by the officers of the Shannon, the instrument needed to perforate the Scotia, and the power to pierce a steamer's hull.

"In essence, the narwhale is armed with a sort of ivory sword, or lance, as certain naturalists have expressed it. It's a king-sized tooth as hard as steel. Some of these teeth have been found buried in the bodies of baleen whales, which the narwhale attacks with invariable success. Others have been wrenched, not without difficulty, from the undersides of vessels that narwhales have pierced clean through, as a gimlet pierces a wine barrel. The museum at the Faculty of Medicine in Paris owns one of these tusks with a length of 2.25 meters and a width at its base of forty-eight centimeters!

"All right then! Imagine this weapon to be ten times stronger and the animal ten times more powerful, launch it at a speed of twenty miles per hour, multiply its mass times its velocity, and you get just the collision we need to cause the specified catastrophe.

"So, until information becomes more abundant, I plump for a sea unicorn of colossal dimensions, no longer armed with a mere lance but with an actual spur, like ironclad frigates or those warships called 'rams,' whose mass and motor power it would possess simultaneously.

"This inexplicable phenomenon is thus explained away—unless it's something else entirely, which, despite everything that has been sighted, studied, explored and experienced, is still possible!"

These last words were cowardly of me; but as far as I could, I wanted to protect my professorial dignity and not lay myself open to laughter from the Americans, who when they do laugh, laugh raucously. I had left myself a loophole. Yet deep down, I had accepted the existence of "the monster."

My article was hotly debated, causing a fine old uproar. It rallied a number of supporters. Moreover, the solution it proposed allowed for free play of the imagination. The human mind enjoys impressive visions of unearthly creatures. Now then, the sea is precisely their best medium, the only setting suitable for the breeding and growing of such giants—next to which such land animals as elephants or rhinoceroses are mere dwarves. The liquid masses

support the largest known species of mammals and perhaps conceal mollusks of incomparable size or crustaceans too frightful to contemplate, such as 100-meter lobsters or crabs weighing 200 metric tons! Why not? Formerly, in prehistoric days, land animals (quadrupeds, apes, reptiles, birds) were built on a gigantic scale. Our Creator cast them using a colossal mold that time has gradually made smaller. With its untold depths, couldn't the sea keep alive such huge specimens of life from another age, this sea that never changes while the land masses undergo almost continuous alteration? Couldn't the heart of the ocean hide the last-remaining varieties of these titanic species, for whom years are centuries and centuries millennia?

But I mustn't let these fantasies run away with me! Enough of these fairy tales that time has changed for me into harsh realities. I repeat: opinion had crystallized as to the nature of this phenomenon, and the public accepted without argument the existence of a prodigious creature that had nothing in common with the fabled sea serpent.

Yet if some saw it purely as a scientific problem to be solved, more practical people, especially in America and England, were determined to purge the ocean of this daunting monster, to insure the safety of transoceanic travel. The industrial and commercial newspapers dealt with the question chiefly from this viewpoint. The Shipping & Mercantile Gazette, the Lloyd's List, France's Packetboat and Maritime & Colonial Review, all the rags devoted to insurance companies—who threatened to raise their premium rates—were unanimous on this point.

Public opinion being pronounced, the States of the Union were the first in the field. In New York preparations were under way for an expedition designed to chase this narwhale. A high-speed frigate, the Abraham Lincoln, was fitted out for putting to sea as soon as possible. The naval arsenals were unlocked for Commander Farragut, who pressed energetically forward with the arming of his frigate.

But, as it always happens, just when a decision had been made to chase the monster, the monster put in no further appearances. For two months nobody

heard a word about it. Not a single ship encountered it. Apparently the unicorn had gotten wise to these plots being woven around it. People were constantly babbling about the creature, even via the Atlantic Cable! Accordingly, the wags claimed that this slippery rascal had waylaid some passing telegram and was making the most of it.

So the frigate was equipped for a far-off voyage and armed with fearsome fishing gear, but nobody knew where to steer it. And impatience grew until, on June 2, word came that the Tampico, a steamer on the San Francisco line sailing from California to Shanghai, had sighted the animal again, three weeks before in the northerly seas of the Pacific.

This news caused intense excitement. Not even a 24-hour breather was granted to Commander Farragut. His provisions were loaded on board. His coal bunkers were overflowing. Not a crewman was missing from his post. To cast off, he needed only to fire and stoke his furnaces! Half a day's delay would have been unforgivable! But Commander Farragut wanted nothing more than to go forth.

I received a letter three hours before the Abraham Lincoln left its Brooklyn pier;* the letter read as follows:

*Author's Note: A pier is a type of wharf expressly set aside for an individual vessel.

Pierre Aronnax Professor at the Paris Museum Fifth Avenue Hotel New York

Sir:

If you would like to join the expedition on the Abraham Lincoln, the government of the Union will be pleased to regard you as France's representative in this undertaking. Commander Farragut has a cabin at your disposal.

Very cordially yours,

J. B. HOBSON, Secretary of the Navy.

Chapter 3

As Master Wishes

THREE SECONDS before the arrival of J. B. Hobson's letter, I no more dreamed of chasing the unicorn than of trying for the Northwest Passage. Three seconds after reading this letter from the honorable Secretary of the Navy, I understood at last that my true vocation, my sole purpose in life, was to hunt down this disturbing monster and rid the world of it.

Even so, I had just returned from an arduous journey, exhausted and badly needing a rest. I wanted nothing more than to see my country again, my friends, my modest quarters by the Botanical Gardens, my dearly beloved collections! But now nothing could hold me back. I forgot everything else, and without another thought of exhaustion, friends, or collections, I accepted the American government's offer.

"Besides," I mused, "all roads lead home to Europe, and our unicorn may be gracious enough to take me toward the coast of France! That fine animal may even let itself be captured in European seas—as a personal favor to me—and I'll bring back to the Museum of Natural History at least half a meter of its ivory lance!"

But in the meantime I would have to look for this narwhale in the northern Pacific Ocean; which meant returning to France by way of the Antipodes.

"Conseil!" I called in an impatient voice.

Conseil was my manservant. A devoted lad who went with me on all my journeys; a gallant Flemish boy whom I genuinely liked and who returned the compliment; a born stoic, punctilious on principle, habitually hardworking, rarely startled by life's surprises, very skillful with his hands, efficient in his every duty, and despite his having a name that means "counsel," never giving advice—not even the unsolicited kind!

From rubbing shoulders with scientists in our little universe by the Botanical Gardens, the boy had come to know a thing or two. In Conseil I had a seasoned specialist in biological classification, an enthusiast who could run with acrobatic agility up and down the whole ladder of branches, groups, classes, subclasses, orders, families, genera, subgenera, species, and varieties. But there his science came to a halt. Classifying was everything to him, so he knew nothing else. Well versed in the theory of classification, he was poorly versed in its practical application, and I doubt that he could tell a sperm whale from a baleen whale! And yet, what a fine, gallant lad!

For the past ten years, Conseil had gone with me wherever science beckoned. Not once did he comment on the length or the hardships of a journey. Never did he object to buckling up his suitcase for any country whatever, China or the Congo, no matter how far off it was. He went here, there, and everywhere in perfect contentment. Moreover, he enjoyed excellent health that defied all ailments, owned solid muscles, but hadn’t a nerve in him, not a sign of nerves—the mental type, I mean.

The lad was thirty years old, and his age to that of his employer was as fifteen is to twenty. Please forgive me for this underhanded way of admitting I had turned forty.

But Conseil had one flaw. He was a fanatic on formality, and he only addressed me in the third person—to the point where it got tiresome.

“Conseil!” I repeated, while feverishly beginning my preparations for departure.

To be sure, I had confidence in this devoted lad. Ordinarily, I never asked whether or not it suited him to go with me on my journeys; but this time an expedition was at issue that could drag on indefinitely, a hazardous undertaking whose purpose was to hunt an animal that could sink a frigate as easily as a walnut shell! There was good reason to stop and think, even for the world’s most emotionless man. What would Conseil say?

“Conseil!” I called a third time.

Conseil appeared.

“Did master summon me?” he said, entering.

“Yes, my boy. Get my things ready, get yours ready. We’re departing in two hours.”

“As master wishes,” Conseil replied serenely.

“We haven’t a moment to lose. Pack as much into my trunk as you can, my traveling kit, my suits, shirts, and socks, don’t bother counting, just squeeze it all in—and hurry!”

“What about master’s collections?” Conseil ventured to observe.

“We’ll deal with them later.”

“What! The archaeotherium, hyracotherium, oreodonts, cheiropotamus, and master’s other fossil skeletons?”

“The hotel will keep them for us.”

“What about master’s live babirusa?”

“They’ll feed it during our absence. Anyhow, we’ll leave instructions to ship the whole menagerie to France.”

“Then we aren’t returning to Paris?” Conseil asked.

“Yes, we are... certainly...,” I replied evasively, “but after we make a detour.”

“Whatever detour master wishes.”

“Oh, it’s nothing really! A route slightly less direct, that’s all. We’re leaving on the Abraham Lincoln.”

“As master thinks best,” Conseil replied placidly.

“You see, my friend, it’s an issue of the monster, the notorious narwhale. We’re going to rid the seas of it! The author of a two-volume work, in quarto, on The Mysteries of the Great Ocean Depths has no excuse for not setting sail with Commander Farragut. It’s a glorious mission but also a dangerous one! We don’t know where it will take us! These beasts can be quite unpredictable! But we’re going just the same! We have a commander who’s game for anything!”

“What master does, I’ll do,” Conseil replied.

“But think it over, because I don’t want to hide anything from you. This is one of those voyages from which people don’t always come back!”

“As master wishes.”

A quarter of an hour later, our trunks were ready. Conseil did them in a flash, and I was sure the lad hadn’t missed a thing, because he classified shirts and suits as expertly as birds and mammals.

The hotel elevator dropped us off in the main vestibule on the mezzanine. I went down a short stair leading to the ground floor. I settled my bill at that huge counter that was always under siege by a considerable crowd. I left instructions for shipping my containers of stuffed animals and dried plants to Paris, France. I opened a line of credit sufficient to cover the babirusa and, Conseil at my heels, I jumped into a carriage.

For a fare of twenty francs, the vehicle went down Broadway to Union Square, took Fourth Ave. to its junction with Bowery St., turned into Katrin St. and halted at Pier 34. There the Katrin ferry transferred men, horses, and carriage to Brooklyn, that great New York annex located on the left bank of the East River, and in a few minutes we arrived at the wharf next to which the Abraham Lincoln was vomiting torrents of black smoke from its two funnels.

Our baggage was immediately carried to the deck of the frigate. I rushed aboard. I asked for Commander Farragut. One of the sailors led me to the afterdeck, where I stood in the presence of a smart-looking officer who extended his hand to me.

“Professor Pierre Aronnax?” he said to me.

“The same,” I replied. “Commander Farragut?”

“In person. Welcome aboard, professor. Your cabin is waiting for you.”

I bowed, and letting the commander attend to getting under way, I was taken to the cabin that had been set aside for me.

The Abraham Lincoln had been perfectly chosen and fitted out for its new assignment. It was a high-speed frigate furnished with superheating equipment that allowed the tension of its steam to build to seven atmospheres. Under this pressure the Abraham Lincoln reached an average speed of 18.3 miles per

hour, a considerable speed but still not enough to cope with our gigantic cetacean.

The frigate's interior accommodations complemented its nautical virtues. I was well satisfied with my cabin, which was located in the stern and opened into the officers' mess.

"We'll be quite comfortable here," I told Conseil.

"With all due respect to master," Conseil replied, "as comfortable as a hermit crab inside the shell of a whelk."

I left Conseil to the proper stowing of our luggage and climbed on deck to watch the preparations for getting under way.

Just then Commander Farragut was giving orders to cast off the last moorings holding the Abraham Lincoln to its Brooklyn pier. And so if I'd been delayed by a quarter of an hour or even less, the frigate would have gone without me, and I would have missed out on this unearthly, extraordinary, and inconceivable expedition, whose true story might well meet with some skepticism.

But Commander Farragut didn't want to waste a single day, or even a single hour, in making for those seas where the animal had just been sighted. He summoned his engineer.

"Are we up to pressure?" he asked the man.

"Aye, sir," the engineer replied.

"Go ahead, then!" Commander Farragut called.

At this order, which was relayed to the engine by means of a compressed-air device, the mechanics activated the start-up wheel. Steam rushed whistling into the gaping valves. Long horizontal pistons groaned and pushed the tie rods of the drive shaft. The blades of the propeller churned the waves with increasing speed, and the Abraham Lincoln moved out majestically amid a spectator-laden escort of some 100 ferries and tenders.*

*Author's Note: Tenders are small steamboats that assist the big liners.

The wharves of Brooklyn, and every part of New York bordering the East River, were crowded with curiosity seekers. Departing from 500,000 throats,

three cheers burst forth in succession. Thousands of handkerchiefs were waving above these tightly packed masses, hailing the Abraham Lincoln until it reached the waters of the Hudson River, at the tip of the long peninsula that forms New York City.

The frigate then went along the New Jersey coast—the wonderful right bank of this river, all loaded down with country homes—and passed by the forts to salutes from their biggest cannons. The Abraham Lincoln replied by three times lowering and hoisting the American flag, whose thirty-nine stars gleamed from the gaff of the mizzen sail; then, changing speed to take the buoy-marked channel that curved into the inner bay formed by the spit of Sandy Hook, it hugged this sand-covered strip of land where thousands of spectators acclaimed us one more time.

The escort of boats and tenders still followed the frigate and only left us when we came abreast of the lightship, whose two signal lights mark the entrance of the narrows to Upper New York Bay.

Three o’clock then sounded. The harbor pilot went down into his dinghy and rejoined a little schooner waiting for him to leeward. The furnaces were stoked; the propeller churned the waves more swiftly; the frigate skirted the flat, yellow coast of Long Island; and at eight o’clock in the evening, after the lights of Fire Island had vanished into the northwest, we ran at full steam onto the dark waters of the Atlantic.

Chapter 4

Ned Land

COMMANDER FARRAGUT was a good seaman, worthy of the frigate he commanded. His ship and he were one. He was its very soul. On the cetacean

question no doubts arose in his mind, and he didn't allow the animal's existence to be disputed aboard his vessel. He believed in it as certain pious women believe in the leviathan from the Book of Job—out of faith, not reason. The monster existed, and he had vowed to rid the seas of it. The man was a sort of Knight of Rhodes, a latter-day Sir Dieudonné of Gozo, on his way to fight an encounter with the dragon devastating the island. Either Commander Farragut would slay the narwhale, or the narwhale would slay Commander Farragut. No middle of the road for these two.

The ship's officers shared the views of their leader. They could be heard chatting, discussing, arguing, calculating the different chances of an encounter, and observing the vast expanse of the ocean. Voluntary watches from the crosstrees of the topgallant sail were self-imposed by more than one who would have cursed such toil under any other circumstances. As often as the sun swept over its daily arc, the masts were populated with sailors whose feet itched and couldn't hold still on the planking of the deck below! And the Abraham Lincoln's stempost hadn't even cut the suspected waters of the Pacific.

As for the crew, they only wanted to encounter the unicorn, harpoon it, haul it on board, and carve it up. They surveyed the sea with scrupulous care. Besides, Commander Farragut had mentioned that a certain sum of $2,000.00 was waiting for the man who first sighted the animal, be he cabin boy or sailor, mate or officer. I'll let the reader decide whether eyes got proper exercise aboard the Abraham Lincoln.

As for me, I didn't lag behind the others and I yielded to no one my share in these daily observations. Our frigate would have had fivescore good reasons for renaming itself the Argus, after that mythological beast with 100 eyes! The lone rebel among us was Conseil, who seemed utterly uninterested in the question exciting us and was out of step with the general enthusiasm on board.

As I said, Commander Farragut had carefully equipped his ship with all the gear needed to fish for a gigantic cetacean. No whaling vessel could have been better armed. We had every known mechanism, from the hand-hurled harpoon,

to the blunderbuss firing barbed arrows, to the duck gun with exploding bullets. On the forecastle was mounted the latest model breech-loading cannon, very heavy of barrel and narrow of bore, a weapon that would figure in the Universal Exhibition of 1867. Made in America, this valuable instrument could fire a four-kilogram conical projectile an average distance of sixteen kilometers without the least bother.

So the Abraham Lincoln wasn't lacking in means of destruction. But it had better still. It had Ned Land, the King of Harpooners.

Gifted with uncommon manual ability, Ned Land was a Canadian who had no equal in his dangerous trade. Dexterity, coolness, bravery, and cunning were virtues he possessed to a high degree, and it took a truly crafty baleen whale or an exceptionally astute sperm whale to elude the thrusts of his harpoon.

Ned Land was about forty years old. A man of great height—over six English feet—he was powerfully built, serious in manner, not very sociable, sometimes headstrong, and quite ill-tempered when crossed. His looks caught the attention, and above all the strength of his gaze, which gave a unique emphasis to his facial appearance.

Commander Farragut, to my thinking, had made a wise move in hiring on this man. With his eye and his throwing arm, he was worth the whole crew all by himself. I can do no better than to compare him with a powerful telescope that could double as a cannon always ready to fire.

To say Canadian is to say French, and as unsociable as Ned Land was, I must admit he took a definite liking to me. No doubt it was my nationality that attracted him. It was an opportunity for him to speak, and for me to hear, that old Rabelaisian dialect still used in some Canadian provinces. The harpooner's family originated in Quebec, and they were already a line of bold fishermen back in the days when this town still belonged to France.

Little by little Ned developed a taste for chatting, and I loved hearing the tales of his adventures in the polar seas. He described his fishing trips and his battles with great natural lyricism. His tales took on the form of an epic poem,

and I felt I was hearing some Canadian Homer reciting his Iliad of the High Arctic regions.

I'm writing of this bold companion as I currently know him. Because we've become old friends, united in that permanent comradeship born and cemented during only the most frightful crises! Ah, my gallant Ned! I ask only to live 100 years more, the longer to remember you!

And now, what were Ned Land's views on this question of a marine monster? I must admit that he flatly didn't believe in the unicorn, and alone on board, he didn't share the general conviction. He avoided even dealing with the subject, for which one day I felt compelled to take him to task.

During the magnificent evening of June 25—in other words, three weeks after our departure—the frigate lay abreast of Cabo Blanco, thirty miles to leeward of the coast of Patagonia. We had crossed the Tropic of Capricorn, and the Strait of Magellan opened less than 700 miles to the south. Before eight days were out, the Abraham Lincoln would plow the waves of the Pacific.

Seated on the afterdeck, Ned Land and I chatted about one thing and another, staring at that mysterious sea whose depths to this day are beyond the reach of human eyes. Quite naturally, I led our conversation around to the giant unicorn, and I weighed our expedition's various chances for success or failure. Then, seeing that Ned just let me talk without saying much himself, I pressed him more closely.

"Ned," I asked him, "how can you still doubt the reality of this cetacean we're after? Do you have any particular reasons for being so skeptical?"

The harpooner stared at me awhile before replying, slapped his broad forehead in one of his standard gestures, closed his eyes as if to collect himself, and finally said:

"Just maybe, Professor Aronnax."

"But Ned, you're a professional whaler, a man familiar with all the great marine mammals—your mind should easily accept this hypothesis of an enormous cetacean, and you ought to be the last one to doubt it under these circumstances!"

“That’s just where you’re mistaken, professor,” Ned replied. “The common man may still believe in fabulous comets crossing outer space, or in prehistoric monsters living at the earth’s core, but astronomers and geologists don’t swallow such fairy tales. It’s the same with whalers. I’ve chased plenty of cetaceans, I’ve harpooned a good number, I’ve killed several. But no matter how powerful and well armed they were, neither their tails or their tusks could puncture the sheet-iron plates of a steamer.”

“Even so, Ned, people mention vessels that narwhale tusks have run clean through.”

“Wooden ships maybe,” the Canadian replied. “But I’ve never seen the like. So till I have proof to the contrary, I’ll deny that baleen whales, sperm whales, or unicorns can do any such thing.”

“Listen to me, Ned—”

“No, no, professor. I’ll go along with anything you want except that. Some gigantic devilfish maybe...?”

“Even less likely, Ned. The devilfish is merely a mollusk, and even this name hints at its semiliquid flesh, because it’s Latin meaning soft one. The devilfish doesn’t belong to the vertebrate branch, and even if it were 500 feet long, it would still be utterly harmless to ships like the Scotia or the Abraham Lincoln. Consequently, the feats of krakens or other monsters of that ilk must be relegated to the realm of fiction.”

“So, Mr. Naturalist,” Ned Land continued in a bantering tone, “you’ll just keep on believing in the existence of some enormous cetacean...?”

“Yes, Ned, I repeat it with a conviction backed by factual logic. I believe in the existence of a mammal with a powerful constitution, belonging to the vertebrate branch like baleen whales, sperm whales, or dolphins, and armed with a tusk made of horn that has tremendous penetrating power.”

“Humph!” the harpooner put in, shaking his head with the attitude of a man who doesn’t want to be convinced.

“Note well, my fine Canadian,” I went on, “if such an animal exists, if it lives deep in the ocean, if it frequents the liquid strata located miles beneath

the surface of the water, it needs to have a constitution so solid, it defies all comparison."

"And why this powerful constitution?" Ned asked.

"Because it takes incalculable strength just to live in those deep strata and withstand their pressure."

"Oh really?" Ned said, tipping me a wink.

"Oh really, and I can prove it to you with a few simple figures."

"Bosh!" Ned replied. "You can make figures do anything you want!"

"In business, Ned, but not in mathematics. Listen to me. Let's accept that the pressure of one atmosphere is represented by the pressure of a column of water thirty-two feet high. In reality, such a column of water wouldn't be quite so high because here we're dealing with salt water, which is denser than fresh water. Well then, when you dive under the waves, Ned, for every thirty-two feet of water above you, your body is tolerating the pressure of one more atmosphere, in other words, one more kilogram per each square centimeter on your body's surface. So it follows that at 320 feet down, this pressure is equal to ten atmospheres, to 100 atmospheres at 3,200 feet, and to 1,000 atmospheres at 32,000 feet, that is, at about two and a half vertical leagues down. Which is tantamount to saying that if you could reach such a depth in the ocean, each square centimeter on your body's surface would be experiencing 1,000 kilograms of pressure. Now, my gallant Ned, do you know how many square centimeters you have on your bodily surface?"

"I haven't the foggiest notion, Professor Aronnax."

"About 17,000."

"As many as that?"

"Yes, and since the atmosphere's pressure actually weighs slightly more than one kilogram per square centimeter, your 17,000 square centimeters are tolerating 17,568 kilograms at this very moment."

"Without my noticing it?"

"Without your noticing it. And if you aren't crushed by so much pressure, it's because the air penetrates the interior of your body with equal pressure.

When the inside and outside pressures are in perfect balance, they neutralize each other and allow you to tolerate them without discomfort. But in the water it's another story."

"Yes, I see," Ned replied, growing more interested. "Because the water surrounds me but doesn't penetrate me."

"Precisely, Ned. So at thirty-two feet beneath the surface of the sea, you'll undergo a pressure of 17,568 kilograms; at 320 feet, or ten times greater pressure, it's 175,680 kilograms; at 3,200 feet, or 100 times greater pressure, it's 1,756,800 kilograms; finally, at 32,000 feet, or 1,000 times greater pressure, it's 17,568,000 kilograms; in other words, you'd be squashed as flat as if you'd just been yanked from between the plates of a hydraulic press!"

"Fire and brimstone!" Ned put in.

"All right then, my fine harpooner, if vertebrates several hundred meters long and proportionate in bulk live at such depths, their surface areas make up millions of square centimeters, and the pressure they undergo must be assessed in billions of kilograms. Calculate, then, how much resistance of bone structure and strength of constitution they'd need in order to withstand such pressures!"

"They'd need to be manufactured," Ned Land replied, "from sheet-iron plates eight inches thick, like ironclad frigates."

"Right, Ned, and then picture the damage such a mass could inflict if it were launched with the speed of an express train against a ship's hull."

"Yes... indeed... maybe," the Canadian replied, staggered by these figures but still not willing to give in.

"Well, have I convinced you?"

"You've convinced me of one thing, Mr. Naturalist. That deep in the sea, such animals would need to be just as strong as you say—if they exist."

"But if they don't exist, my stubborn harpooner, how do you explain the accident that happened to the Scotia?"

"It's maybe...," Ned said, hesitating.

"Go on!"

“Because... it just couldn’t be true!” the Canadian replied, unconsciously echoing a famous catchphrase of the scientist Arago.

But this reply proved nothing, other than how bullheaded the harpooner could be. That day I pressed him no further. The Scotia’s accident was undeniable. Its hole was real enough that it had to be plugged up, and I don’t think a hole’s existence can be more emphatically proven. Now then, this hole didn’t make itself, and since it hadn’t resulted from underwater rocks or underwater machines, it must have been caused by the perforating tool of some animal.

Now, for all the reasons put forward to this point, I believed that this animal was a member of the branch Vertebrata, class Mammalia, group Pisciforma, and finally, order Cetacea. As for the family in which it would be placed (baleen whale, sperm whale, or dolphin), the genus to which it belonged, and the species in which it would find its proper home, these questions had to be left for later. To answer them called for dissecting this unknown monster; to dissect it called for catching it; to catch it called for harpooning it—which was Ned Land’s business; to harpoon it called for sighting it—which was the crew’s business; and to sight it called for encountering it—which was a chancy business.

Chapter 5

At Random!

FOR SOME WHILE the voyage of the Abraham Lincoln was marked by no incident. But one circumstance arose that displayed Ned Land’s marvelous skills and showed just how much confidence we could place in him.

Off the Falkland Islands on June 30, the frigate came in contact with a fleet of American whalers, and we learned that they hadn't seen the narwhale. But one of them, the captain of the Monroe, knew that Ned Land had shipped aboard the Abraham Lincoln and asked his help in hunting a baleen whale that was in sight. Anxious to see Ned Land at work, Commander Farragut authorized him to make his way aboard the Monroe. And the Canadian had such good luck that with a right-and-left shot, he harpooned not one whale but two, striking the first straight to the heart and catching the other after a few minutes' chase!

Assuredly, if the monster ever had to deal with Ned Land's harpoon, I wouldn't bet on the monster.

The frigate sailed along the east coast of South America with prodigious speed. By July 3 we were at the entrance to the Strait of Magellan, abreast of Cabo de las Virgenes. But Commander Farragut was unwilling to attempt this tortuous passageway and maneuvered instead to double Cape Horn.

The crew sided with him unanimously. Indeed, were we likely to encounter the narwhale in such a cramped strait? Many of our sailors swore that the monster couldn't negotiate this passageway simply because "he's too big for it!"

Near three o'clock in the afternoon on July 6, fifteen miles south of shore, the Abraham Lincoln doubled that solitary islet at the tip of the South American continent, that stray rock Dutch seamen had named Cape Horn after their hometown of Hoorn. Our course was set for the northwest, and the next day our frigate's propeller finally churned the waters of the Pacific.

"Open your eyes! Open your eyes!" repeated the sailors of the Abraham Lincoln.

And they opened amazingly wide. Eyes and spyglasses (a bit dazzled, it is true, by the vista of $2,000.00) didn't remain at rest for an instant. Day and night we observed the surface of the ocean, and those with nyctalopic eyes, whose ability to see in the dark increased their chances by fifty percent, had an excellent shot at winning the prize.

As for me, I was hardly drawn by the lure of money and yet was far from the least attentive on board. Snatching only a few minutes for meals and a few hours for sleep, come rain or come shine, I no longer left the ship's deck. Sometimes bending over the forecastle railings, sometimes leaning against the sternrail, I eagerly scoured that cotton-colored wake that whitened the ocean as far as the eye could see! And how many times I shared the excitement of general staff and crew when some unpredictable whale lifted its blackish back above the waves. In an instant the frigate's deck would become densely populated. The cowls over the companionways would vomit a torrent of sailors and officers. With panting chests and anxious eyes, we each would observe the cetacean's movements. I stared; I stared until I nearly went blind from a worn-out retina, while Conseil, as stoic as ever, kept repeating to me in a calm tone:

"If master's eyes would kindly stop bulging, master will see farther!"

But what a waste of energy! The Abraham Lincoln would change course and race after the animal sighted, only to find an ordinary baleen whale or a common sperm whale that soon disappeared amid a chorus of curses!

However, the weather held good. Our voyage was proceeding under the most favorable conditions. By then it was the bad season in these southernmost regions, because July in this zone corresponds to our January in Europe; but the sea remained smooth and easily visible over a vast perimeter.

Ned Land still kept up the most tenacious skepticism; beyond his spells on watch, he pretended that he never even looked at the surface of the waves, at least while no whales were in sight. And yet the marvelous power of his vision could have performed yeoman service. But this stubborn Canadian spent eight hours out of every twelve reading or sleeping in his cabin. A hundred times I chided him for his unconcern.

"Bah!" he replied. "Nothing's out there, Professor Aronnax, and if there is some animal, what chance would we have of spotting it? Can't you see we're just wandering around at random? People say they've sighted this slippery beast again in the Pacific high seas—I'm truly willing to believe it, but two

months have already gone by since then, and judging by your narwhale's personality, it hates growing moldy from hanging out too long in the same waterways! It's blessed with a terrific gift for getting around. Now, professor, you know even better than I that nature doesn't violate good sense, and she wouldn't give some naturally slow animal the ability to move swiftly if it hadn't a need to use that talent. So if the beast does exist, it's already long gone!"

I had no reply to this. Obviously we were just groping blindly. But how else could we go about it? All the same, our chances were automatically pretty limited. Yet everyone still felt confident of success, and not a sailor on board would have bet against the narwhale appearing, and soon.

On July 20 we cut the Tropic of Capricorn at longitude 105 degrees, and by the 27th of the same month, we had cleared the equator on the 110th meridian. These bearings determined, the frigate took a more decisive westward heading and tackled the seas of the central Pacific. Commander Farragut felt, and with good reason, that it was best to stay in deep waters and keep his distance from continents or islands, whose neighborhoods the animal always seemed to avoid—"No doubt," our bosun said, "because there isn't enough water for him!" So the frigate kept well out when passing the Tuamotu, Marquesas, and Hawaiian Islands, then cut the Tropic of Cancer at longitude 132 degrees and headed for the seas of China.

We were finally in the area of the monster's latest antics! And in all honesty, shipboard conditions became life-threatening. Hearts were pounding hideously, gearing up for futures full of incurable aneurysms. The entire crew suffered from a nervous excitement that it's beyond me to describe. Nobody ate, nobody slept. Twenty times a day some error in perception, or the optical illusions of some sailor perched in the crosstrees, would cause intolerable anguish, and this emotion, repeated twenty times over, kept us in a state of irritability so intense that a reaction was bound to follow.

And this reaction wasn't long in coming. For three months, during which each day seemed like a century, the Abraham Lincoln plowed all the northerly

seas of the Pacific, racing after whales sighted, abruptly veering off course, swerving sharply from one tack to another, stopping suddenly, putting on steam and reversing engines in quick succession, at the risk of stripping its gears, and it didn't leave a single point unexplored from the beaches of Japan to the coasts of America. And we found nothing! Nothing except an immenseness of deserted waves! Nothing remotely resembling a gigantic narwhale, or an underwater islet, or a derelict shipwreck, or a runaway reef, or anything the least bit unearthly!

So the reaction set in. At first, discouragement took hold of people's minds, opening the door to disbelief. A new feeling appeared on board, made up of three-tenths shame and seven-tenths fury. The crew called themselves "out-and-out fools" for being hoodwinked by a fairy tale, then grew steadily more furious! The mountains of arguments amassed over a year collapsed all at once, and each man now wanted only to catch up on his eating and sleeping, to make up for the time he had so stupidly sacrificed.

With typical human fickleness, they jumped from one extreme to the other. Inevitably, the most enthusiastic supporters of the undertaking became its most energetic opponents. This reaction mounted upward from the bowels of the ship, from the quarters of the bunker hands to the messroom of the general staff; and for certain, if it hadn't been for Commander Farragut's characteristic stubbornness, the frigate would ultimately have put back to that cape in the south.

But this futile search couldn't drag on much longer. The Abraham Lincoln had done everything it could to succeed and had no reason to blame itself. Never had the crew of an American naval craft shown more patience and zeal; they weren't responsible for this failure; there was nothing to do but go home.

A request to this effect was presented to the commander. The commander stood his ground. His sailors couldn't hide their discontent, and their work suffered because of it. I'm unwilling to say that there was mutiny on board, but after a reasonable period of intransigence, Commander Farragut, like Christopher Columbus before him, asked for a grace period of just three days

more. After this three-day delay, if the monster hadn't appeared, our helmsman would give three turns of the wheel, and the Abraham Lincoln would chart a course toward European seas.

This promise was given on November 2. It had the immediate effect of reviving the crew's failing spirits. The ocean was observed with renewed care. Each man wanted one last look with which to sum up his experience. Spyglasses functioned with feverish energy. A supreme challenge had been issued to the giant narwhale, and the latter had no acceptable excuse for ignoring this Summons to Appear!

Two days passed. The Abraham Lincoln stayed at half steam. On the offchance that the animal might be found in these waterways, a thousand methods were used to spark its interest or rouse it from its apathy. Enormous sides of bacon were trailed in our wake, to the great satisfaction, I must say, of assorted sharks. While the Abraham Lincoln heaved to, its longboats radiated in every direction around it and didn't leave a single point of the sea unexplored. But the evening of November 4 arrived with this underwater mystery still unsolved.

At noon the next day, November 5, the agreed-upon delay expired. After a position fix, true to his promise, Commander Farragut would have to set his course for the southeast and leave the northerly regions of the Pacific decisively behind.

By then the frigate lay in latitude 31 degrees 15′ north and longitude 136 degrees 42′ east. The shores of Japan were less than 200 miles to our leeward. Night was coming on. Eight o'clock had just struck. Huge clouds covered the moon's disk, then in its first quarter. The sea undulated placidly beneath the frigate's stempost.

Just then I was in the bow, leaning over the starboard rail. Conseil, stationed beside me, stared straight ahead. Roosting in the shrouds, the crew examined the horizon, which shrank and darkened little by little. Officers were probing the increasing gloom with their night glasses. Sometimes the murky

ocean sparkled beneath moonbeams that darted between the fringes of two clouds. Then all traces of light vanished into the darkness.

Observing Conseil, I discovered that, just barely, the gallant lad had fallen under the general influence. At least so I thought. Perhaps his nerves were twitching with curiosity for the first time in history.

“Come on, Conseil!” I told him. “Here’s your last chance to pocket that $2,000.00!”

“If master will permit my saying so,” Conseil replied, “I never expected to win that prize, and the Union government could have promised $100,000.00 and been none the poorer.”

“You’re right, Conseil, it turned out to be a foolish business after all, and we jumped into it too hastily. What a waste of time, what a futile expense of emotion! Six months ago we could have been back in France—”

“In master’s little apartment,” Conseil answered. “In master’s museum! And by now I would have classified master’s fossils. And master’s babirusa would be ensconced in its cage at the zoo in the Botanical Gardens, and it would have attracted every curiosity seeker in town!”

“Quite so, Conseil, and what’s more, I imagine that people will soon be poking fun at us!”

“To be sure,” Conseil replied serenely, “I do think they’ll have fun at master’s expense. And must it be said...?”

“It must be said, Conseil.”

“Well then, it will serve master right!”

“How true!”

“When one has the honor of being an expert as master is, one mustn’t lay himself open to—”

Conseil didn’t have time to complete the compliment. In the midst of the general silence, a voice became audible. It was Ned Land’s voice, and it shouted:

“Ahoy! There’s the thing in question, abreast of us to leeward!”

Chapter 6

At Full Steam

AT THIS SHOUT the entire crew rushed toward the harpooner—commander, officers, mates, sailors, cabin boys, down to engineers leaving their machinery and stokers neglecting their furnaces. The order was given to stop, and the frigate merely coasted.

By then the darkness was profound, and as good as the Canadian's eyes were, I still wondered how he could see—and what he had seen. My heart was pounding fit to burst.

But Ned Land was not mistaken, and we all spotted the object his hand was indicating.

Two cable lengths off the Abraham Lincoln's starboard quarter, the sea seemed to be lit up from underneath. This was no mere phosphorescent phenomenon, that much was unmistakable. Submerged some fathoms below the surface of the water, the monster gave off that very intense but inexplicable glow that several captains had mentioned in their reports. This magnificent radiance had to come from some force with a great illuminating capacity. The edge of its light swept over the sea in an immense, highly elongated oval, condensing at the center into a blazing core whose unbearable glow diminished by degrees outward.

"It's only a cluster of phosphorescent particles!" exclaimed one of the officers.

"No, sir," I answered with conviction. "Not even angel-wing clams or salps have ever given off such a powerful light. That glow is basically electric in nature. Besides... look, look! It's shifting! It's moving back and forth! It's darting at us!"

A universal shout went up from the frigate.

“Quiet!” Commander Farragut said. “Helm hard to leeward! Reverse engines!”

Sailors rushed to the helm, engineers to their machinery. Under reverse steam immediately, the Abraham Lincoln beat to port, sweeping in a semicircle.

“Right your helm! Engines forward!” Commander Farragut called.

These orders were executed, and the frigate swiftly retreated from this core of light.

My mistake. It wanted to retreat, but the unearthly animal came at us with a speed double our own.

We gasped. More stunned than afraid, we stood mute and motionless. The animal caught up with us, played with us. It made a full circle around the frigate—then doing fourteen knots—and wrapped us in sheets of electricity that were like luminous dust. Then it retreated two or three miles, leaving a phosphorescent trail comparable to those swirls of steam that shoot behind the locomotive of an express train. Suddenly, all the way from the dark horizon where it had gone to gather momentum, the monster abruptly dashed toward the Abraham Lincoln with frightening speed, stopped sharply twenty feet from our side plates, and died out—not by diving under the water, since its glow did not recede gradually—but all at once, as if the source of this brilliant emanation had suddenly dried up. Then it reappeared on the other side of the ship, either by circling around us or by gliding under our hull. At any instant a collision could have occurred that would have been fatal to us.

Meanwhile I was astonished at the frigate’s maneuvers. It was fleeing, not fighting. Built to pursue, it was being pursued, and I commented on this to Commander Farragut. His face, ordinarily so emotionless, was stamped with indescribable astonishment.

“Professor Aronnax,” he answered me, “I don’t know what kind of fearsome creature I’m up against, and I don’t want my frigate running foolish risks in all this darkness. Besides, how should we attack this unknown

creature, how should we defend ourselves against it? Let's wait for daylight, and then we'll play a different role."

"You've no further doubts, commander, as to the nature of this animal?"

"No, sir, it's apparently a gigantic narwhale, and an electric one to boot."

"Maybe," I added, "it's no more approachable than an electric eel or an electric ray!"

"Right," the commander replied. "And if it has their power to electrocute, it's surely the most dreadful animal ever conceived by our Creator. That's why I'll keep on my guard, sir."

The whole crew stayed on their feet all night long. No one even thought of sleeping. Unable to compete with the monster's speed, the Abraham Lincoln slowed down and stayed at half steam. For its part, the narwhale mimicked the frigate, simply rode with the waves, and seemed determined not to forsake the field of battle.

However, near midnight it disappeared, or to use a more appropriate expression, "it went out," like a huge glowworm. Had it fled from us? We were duty bound to fear so rather than hope so. But at 12:53 in the morning, a deafening hiss became audible, resembling the sound made by a waterspout expelled with tremendous intensity.

By then Commander Farragut, Ned Land, and I were on the afterdeck, peering eagerly into the profound gloom.

"Ned Land," the commander asked, "you've often heard whales bellowing?"

"Often, sir, but never a whale like this, whose sighting earned me $2,000.00."

"Correct, the prize is rightfully yours. But tell me, isn't that the noise cetaceans make when they spurt water from their blowholes?"

"The very noise, sir, but this one's way louder. So there can be no mistake. There's definitely a whale lurking in our waters. With your permission, sir," the harpooner added, "tomorrow at daybreak we'll have words with it."

“If it’s in a mood to listen to you, Mr. Land,” I replied in a tone far from convinced.

“Let me get within four harpoon lengths of it,” the Canadian shot back, “and it had better listen!”

“But to get near it,” the commander went on, “I’d have to put a whaleboat at your disposal?”

“Certainly, sir.”

“That would be gambling with the lives of my men.”

“And with my own!” the harpooner replied simply.

Near two o’clock in the morning, the core of light reappeared, no less intense, five miles to windward of the Abraham Lincoln. Despite the distance, despite the noise of wind and sea, we could distinctly hear the fearsome thrashings of the animal’s tail, and even its panting breath. Seemingly, the moment this enormous narwhale came up to breathe at the surface of the ocean, air was sucked into its lungs like steam into the huge cylinders of a 2,000-horsepower engine.

“Hmm!” I said to myself. “A cetacean as powerful as a whole cavalry regiment—now that’s a whale of a whale!”

We stayed on the alert until daylight, getting ready for action. Whaling gear was set up along the railings. Our chief officer loaded the blunderbusses, which can launch harpoons as far as a mile, and long duck guns with exploding bullets that can mortally wound even the most powerful animals. Ned Land was content to sharpen his harpoon, a dreadful weapon in his hands.

At six o’clock day began to break, and with the dawn’s early light, the narwhale’s electric glow disappeared. At seven o’clock the day was well along, but a very dense morning mist shrank the horizon, and our best spyglasses were unable to pierce it. The outcome: disappointment and anger.

I hoisted myself up to the crosstrees of the mizzen sail. Some officers were already perched on the mastheads.

At eight o’clock the mist rolled ponderously over the waves, and its huge curls were lifting little by little. The horizon grew wider and clearer all at once.

Suddenly, just as on the previous evening, Ned Land's voice was audible.

"There's the thing in question, astern to port!" the harpooner shouted.

Every eye looked toward the point indicated.

There, a mile and a half from the frigate, a long blackish body emerged a meter above the waves. Quivering violently, its tail was creating a considerable eddy. Never had caudal equipment thrashed the sea with such power. An immense wake of glowing whiteness marked the animal's track, sweeping in a long curve.

Our frigate drew nearer to the cetacean. I examined it with a completely open mind. Those reports from the Shannon and the Helvetia had slightly exaggerated its dimensions, and I put its length at only 250 feet. Its girth was more difficult to judge, but all in all, the animal seemed to be wonderfully proportioned in all three dimensions.

While I was observing this phenomenal creature, two jets of steam and water sprang from its blowholes and rose to an altitude of forty meters, which settled for me its mode of breathing. From this I finally concluded that it belonged to the branch Vertebrata, class Mammalia, subclass Monodelphia, group Pisciforma, order Cetacea, family... but here I couldn't make up my mind. The order Cetacea consists of three families, baleen whales, sperm whales, dolphins, and it's in this last group that narwhales are placed. Each of these families is divided into several genera, each genus into species, each species into varieties. So I was still missing variety, species, genus, and family, but no doubt I would complete my classifying with the aid of Heaven and Commander Farragut.

The crew were waiting impatiently for orders from their leader. The latter, after carefully observing the animal, called for his engineer. The engineer raced over.

"Sir," the commander said, "are you up to pressure?"

"Aye, sir," the engineer replied.

"Fine. Stoke your furnaces and clap on full steam!"

Three cheers greeted this order. The hour of battle had sounded. A few moments later, the frigate's two funnels vomited torrents of black smoke, and its deck quaked from the trembling of its boilers.

Driven forward by its powerful propeller, the Abraham Lincoln headed straight for the animal. Unconcerned, the latter let us come within half a cable length; then, not bothering to dive, it got up a little speed, retreated, and was content to keep its distance.

This chase dragged on for about three-quarters of an hour without the frigate gaining two fathoms on the cetacean. At this rate, it was obvious that we would never catch up with it.

Infuriated, Commander Farragut kept twisting the thick tuft of hair that flourished below his chin.

"Ned Land!" he called.

The Canadian reported at once.

"Well, Mr. Land," the commander asked, "do you still advise putting my longboats to sea?"

"No, sir," Ned Land replied, "because that beast won't be caught against its will."

"Then what should we do?"

"Stoke up more steam, sir, if you can. As for me, with your permission I'll go perch on the bobstays under the bowsprit, and if we can get within a harpoon length, I'll harpoon the brute."

"Go to it, Ned," Commander Farragut replied. "Engineer," he called, "keep the pressure mounting!"

Ned Land made his way to his post. The furnaces were urged into greater activity; our propeller did forty-three revolutions per minute, and steam shot from the valves. Heaving the log, we verified that the Abraham Lincoln was going at the rate of 18.5 miles per hour.

But that damned animal also did a speed of 18.5.

For the next hour our frigate kept up this pace without gaining a fathom! This was humiliating for one of the fastest racers in the American navy. The

crew were working up into a blind rage. Sailor after sailor heaved insults at the monster, which couldn't be bothered with answering back. Commander Farragut was no longer content simply to twist his goatee; he chewed on it.

The engineer was summoned once again.

"You're up to maximum pressure?" the commander asked him.

"Aye, sir," the engineer replied.

"And your valves are charged to...?"

"To six and a half atmospheres."

"Charge them to ten atmospheres."

A typical American order if I ever heard one. It would have sounded just fine during some Mississippi paddle-wheeler race, to "outstrip the competition!"

"Conseil," I said to my gallant servant, now at my side, "you realize that we'll probably blow ourselves skyhigh?"

"As master wishes!" Conseil replied.

All right, I admit it: I did wish to run this risk!

The valves were charged. More coal was swallowed by the furnaces. Ventilators shot torrents of air over the braziers. The Abraham Lincoln's speed increased. Its masts trembled down to their blocks, and swirls of smoke could barely squeeze through the narrow funnels.

We heaved the log a second time.

"Well, helmsman?" Commander Farragut asked.

"19.3 miles per hour, sir."

"Keep stoking the furnaces."

The engineer did so. The pressure gauge marked ten atmospheres. But no doubt the cetacean itself had "warmed up," because without the least trouble, it also did 19.3.

What a chase! No, I can't describe the excitement that shook my very being. Ned Land stayed at his post, harpoon in hand. Several times the animal let us approach.

"We're overhauling it!" the Canadian would shout.

Then, just as he was about to strike, the cetacean would steal off with a swiftness I could estimate at no less than thirty miles per hour. And even at our maximum speed, it took the liberty of thumbing its nose at the frigate by running a full circle around us! A howl of fury burst from every throat!

By noon we were no farther along than at eight o'clock in the morning.

Commander Farragut then decided to use more direct methods.

"Bah!" he said. "So that animal is faster than the Abraham Lincoln. All right, we'll see if it can outrun our conical shells! Mate, man the gun in the bow!"

Our forecastle cannon was immediately loaded and leveled. The cannoneer fired a shot, but his shell passed some feet above the cetacean, which stayed half a mile off.

"Over to somebody with better aim!" the commander shouted. "And $500.00 to the man who can pierce that infernal beast!"

Calm of eye, cool of feature, an old gray-bearded gunner—I can see him to this day—approached the cannon, put it in position, and took aim for a good while. There was a mighty explosion, mingled with cheers from the crew.

The shell reached its target; it hit the animal, but not in the usual fashion—it bounced off that rounded surface and vanished into the sea two miles out.

"Oh drat!" said the old gunner in his anger. "That rascal must be covered with six-inch armor plate!"

"Curse the beast!" Commander Farragut shouted.

The hunt was on again, and Commander Farragut leaned over to me, saying:

"I'll chase that animal till my frigate explodes!"

"Yes," I replied, "and nobody would blame you!"

We could still hope that the animal would tire out and not be as insensitive to exhaustion as our steam engines. But no such luck. Hour after hour went by without it showing the least sign of weariness.

However, to the Abraham Lincoln's credit, it must be said that we struggled on with tireless persistence. I estimate that we covered a distance of

at least 500 kilometers during this ill-fated day of November 6. But night fell and wrapped the surging ocean in its shadows.

By then I thought our expedition had come to an end, that we would never see this fantastic animal again. I was mistaken.

At 10:50 in the evening, that electric light reappeared three miles to windward of the frigate, just as clear and intense as the night before.

The narwhale seemed motionless. Was it asleep perhaps, weary from its workday, just riding with the waves? This was our chance, and Commander Farragut was determined to take full advantage of it.

He gave his orders. The Abraham Lincoln stayed at half steam, advancing cautiously so as not to awaken its adversary. In midocean it's not unusual to encounter whales so sound asleep they can successfully be attacked, and Ned Land had harpooned more than one in its slumber. The Canadian went to resume his post on the bobstays under the bowsprit.

The frigate approached without making a sound, stopped two cable lengths from the animal and coasted. Not a soul breathed on board. A profound silence reigned over the deck. We were not 100 feet from the blazing core of light, whose glow grew stronger and dazzled the eyes.

Just then, leaning over the forecastle railing, I saw Ned Land below me, one hand grasping the martingale, the other brandishing his dreadful harpoon. Barely twenty feet separated him from the motionless animal.

All at once his arm shot forward and the harpoon was launched. I heard the weapon collide resonantly, as if it had hit some hard substance.

The electric light suddenly went out, and two enormous waterspouts crashed onto the deck of the frigate, racing like a torrent from stem to stern, toppling crewmen, breaking spare masts and yardarms from their lashings.

A hideous collision occurred, and thrown over the rail with no time to catch hold of it, I was hurled into the sea.